

Rabeya Khan

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PROFILE

Motivated Electrical and Electronic Engineering graduate with a strong academic and research background in renewable energy, machine learning, and computational modeling. Skilled in technical communication, scientific writing, and mentoring, with experience supporting students in research and engineering projects. Experienced in funded research projects, IoT, embedded systems, and interdisciplinary engineering. Strong communication, teamwork, and presentation skills, with a long-term goal of building a successful career in teaching and advanced research.

EDUCATION

B.Sc. Engineering in Electrical and Electronic Engineering 01/2020 – 09/2025 | Jamalpur, Bangladesh
Jamalpur Science & Technology University (JSTU)
CGPA: 3.81/4.00 (Class position: 2nd)

PEER-REVIEWED PUBLICATIONS

Device Modeling and Performance Enhancement of Ecologically Viable CsGel₃-Based Perovskite Solar Cells, *Nano Select* 27/04/2025
Rabeya Khan, I. K. Gurusul Ghosh Apurba, Rahat Ul Nasib, Md Rasidul Islam PhD

Numerical simulation and performance enhancement of CsBi₃I₁₀-based heterojunction solar cell, *Solar Energy* 22/03/2025
Rabeya Khan, Nadira Farjana, Mst. Jahida Akter Jim, Jehan Y. Al-Humaidi, Md Rasidul Islam, Md Masud Rana

The Density Function Theory Analysis of Mg₃AsX₃ Perovskite Materials: Strain Effect Study 15/10/2024
Materials Science in Semiconductor Processing
I.K. Gusral Ghosh Apurba, Md. Rasidul Islam, Rabeya Khan, Md Rahat Ul Nasib, Md. Ferdous Rahman, Farhad Ilahi Bakhsh, Abu Zahid, Sohail Ahmad

PROFESSIONAL TRAINING

Certificate in Teaching and Research (CTR) 06/2026 – 08/2026 | Narayanganj, Bangladesh
Center of Excellence for Teaching and Learning, Green University of Bangladesh

- Developed skills in university-level classroom management, lesson planning, student engagement, and effective student interaction.
- Gained practical knowledge of technology-enhanced teaching, including the digitalization of classroom activities and assessment systems.
- Strengthened emotional intelligence, emotional regulation, communication, and interpersonal skills for effective teaching and academic environments.

ACADEMIC AND RESEARCH EXPERIENCE

Teaching Assistant, *Department of EEE, Jamalpur Science & Technology University* 09/2024 – 08/2025 | Jamalpur, Bangladesh

- Delivered foundational lectures and supervised laboratory experiments for undergraduate students in *Communication Engineering Laboratory* courses.
- Guided students through hardware setups, circuit operational analysis, and software tool operations.

Research Assistant, *Robo Tech Valley* 05/2026 – Present | Dhaka, Bangladesh

- Conducting IoT and AI-driven systems modeling; preparing engineering manuscripts for international publications.

Undergraduate Research Assistant, *Supervisor: Dr. Md. Rasidul Islam* 06/2022 – 09/2025

- Executed computational efficiency modeling and first-principles materials simulations of perovskite solar cells using SCAPS-1D and Quantum ESPRESSO.

HIGH-IMPACT CAPSTONE PROJECTS

AquaSense IoT: An Cloud-Integrated Framework for Real-Time Water Quality Monitoring and Control Using pH, DO, TDS, Temperature, and Water Level Sensors 05/2026 – Present
Research Assistant
Developed a two-tier IoT system for real-time water quality monitoring, cloud communication, and automated environmental control using Arduino, ESP32, and Firebase.

Self-Driving Car with Automatic Currency Counter and Barcode Reader for the Blind People, *Team leader* 04/2023 – Present
Leading the development of an autonomous vehicle integrating deep-learning-based navigation, automated currency recognition, and barcode detection to assist visually impaired users.

Bridging the Lab-to-Field Gap: Attention-Consistent Vision Transformers with Faithfulness-Aware Explanation for Robust Rice Disease Diagnosis 07/2026 – 07/2026
Research Assistant
Developed a Vision Transformer-based rice disease detection framework with attention-consistency learning for robust field deployment under domain shift.

A Biogas Power Plant Integrated with IoT-based Monitoring System and Machine Learning-Based analysis 09/2024 – 03/2025
Investigator
Developing an IoT-enabled monitoring system for biogas and electricity generation with real-time data acquisition.

Design, Set up and Analysis of Off-Grid Solar System, *Co-mentor* 13/05/2024 – 26/05/2024
Co-mentored the design and performance analysis of a 5 kW AC/DC off-grid solar power system.

LANGUAGES

Bengali Native	English IELTS Academic: 7.0 overall (no band below 6.0)
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SKILLS

Programming Languages Assembly Language, C, Python, MATLAB, Arduino	Simulation & Analysis Tools SCAPS 1D, OriginPro, Quantum ESPRESSO, Simulink, AutoCAD, LOGO! Soft Comfort	Soft Skills Public speaking, Presentation, Leadership, Teamwork, Communication	Hardware Platforms Raspberry Pi, Arduino, ESP32
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EXTRACURRICULAR ACTIVITIES

President - Jamalpur Science & Technology University Debating Society 05/2025 – 11/2025 | Jamalpur, Bangladesh
Led the JSTU Debating Society by organizing and presiding over university-level debates, workshops, and training programs; represented the university in inter-university competitions, fostering a strong culture of public speaking, critical thinking, and collaborative argumentation while enhancing the society’s reputation.

AWARDS AND HONORS

Merit Scholarship Award 2023 , <i>Jamalpur Science & Technology University (JSTU)</i>	01/12/2024
10th Prize-Grand Finale of Farm Mechanism Idea Competition 2025 , <i>Bangladesh Agricultural Research Institute, Gazipur</i>	01/12/2024
Merit Scholarship Award 2022 , <i>Jamalpur Science & Technology University (JSTU)</i>	01/12/2023
First Position – Innovation Showcasing Programme , <i>E-Governance and Innovation Committee, JSTU</i>	01/12/2023
Merit Scholarship Award 2021 , <i>Jamalpur Science & Technology University (JSTU)</i>	01/12/2022
Champion – Project Showcasing, National Conference on Robotics and IoT, Dept. of CSE, JSTU	01/12/2022
Champion-Project Showcasing, JSTU EEE Day	01/12/2021
Champion - Idea Contest, JSTU EEE Day	01/12/2021
First Position - Public Speaking Competition, JSTU on National Children's Day	01/12/2021
Champion and Best Speaker - DIUDC Bitorko Mohaparbon 1426 (Bengali) <i>Daffodil International University Debating Club (DIUDC)</i>	01/12/2018

SUBMITTED RESEARCH

DFT and Machine Learning Analysis of Eco-friendly Formamidinium Tin Iodide Perovskite Solar Cells: Optimizing Charge Transport Layers through SCAPS-1D Simulations
Rabeya Khan, Md Rasidul Islam, I.K. Gusral Ghosh Apurba, Md. Amimul Ihsan, Rahat Ul Nasib, Mohammad Arefin, Md Rifatul Islam, Abu Zahid

Device modeling and performance analysis of CsGel2Br-based single junction and CsGel2Br/GaAs-based heterojunction perovskite solar cell with appropriate charge transport layers
Rabeya Khan, Mst. Jahida Akter Jim, Nadira Farjana, A. M. Quraishi, Sobhi M. Gomha, Md Rasidul Islam, Md Masud Rana

REFERENCES

Dr. Md. Rasidul Islam, *Assistant Professor, Dept. of EEE, Jamalpur Science & Technology University*
rasidul@jstu.ac.bd, +1 (903) 817-7576

Md. Shizer Rahman, *Chairman, Dept. of EEE, Jamalpur Science & Technology University*
shizer@jstu.ac.bd, 01793781079